

8. Introduction to symmetry-protected topological (SPT) phases

	short-range entangled (SRE)	long-range entangled (LRE)
without symmetry	invertible TO	intrinsic TO (MTC)
with symmetry	SPT	SET (G+MTC) ↑ enriched

Note: Sometimes, are called invertible phases, in the sense that there exist inverse for these phase.

$$\left\{ \begin{array}{c} A \\ A^{-1} \end{array} \right\} \rightarrow I = \text{vac}$$

$\left\{ \begin{array}{l} \text{Topological phases} \rightarrow \text{Abelian monoid under stacking} \\ \text{invertible phases} \rightarrow \text{Abelian group under stacking} \end{array} \right.$

$\downarrow \exists \text{ inverse}$

invertible phase

- invertible TO: Gapped states without anyons, but still can NOT be deformed into trivial product state.
- G-SPT: Gapped states without anyons, but still can NOT be deformed into trivial product state while preserving the symmetry G .

Known classification for invertible TO:

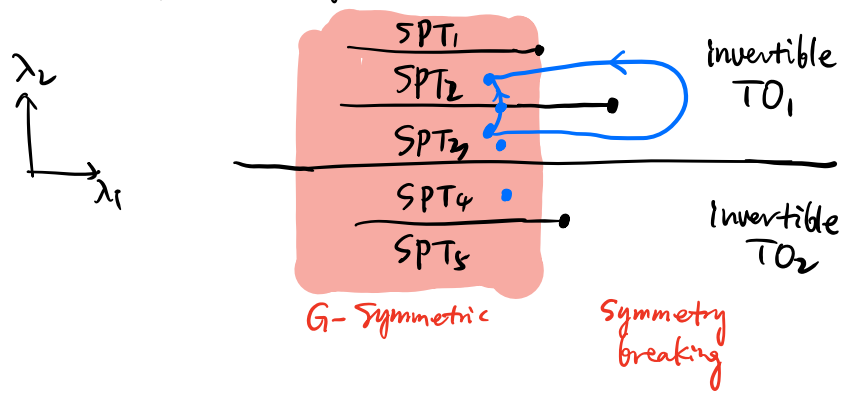
dim	0	1	2	3
bosonic	0	0	$\mathbb{Z}$	0
fermionic	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$	0

$\downarrow$ bosonic/fermionic $\downarrow$ Majorana chain $\downarrow$ P+ip SC

$\rightarrow$ Eg state

Note: fermionic iTO = fermion SPT protected by $G_f = \mathbb{Z}_2^f = \{1, (-1)^F\}$

Schematic phase diagram -



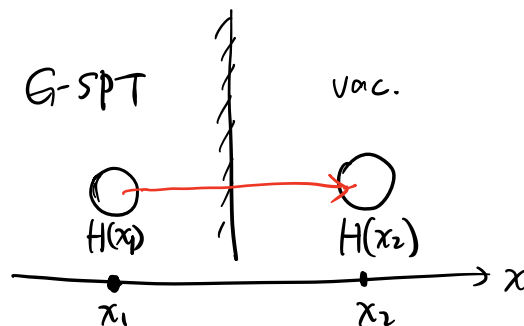
Depending on microscopic dof & (non)interacting:

SPT	noninteracting	interacting
bosonic	/	bSPT → this chapter
fermionic	TI/TSC ↓ last chapter	fSPT → last chapter?

Similar to TI/TSC, nontrivial properties of SPT is on the edge.

Possible edge states for SPT:

- (1) G symmetry breaking
- (2) anomalous SET (with edge GSD)
- (3) gapless
- ~~(4)~~ unique gapped symmetric edge state.

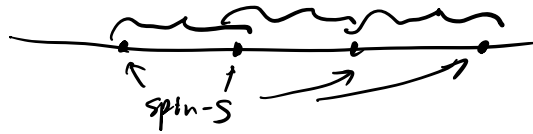


homotopy path $H(x)$: $H(x_1) \rightarrow H(x_2)$

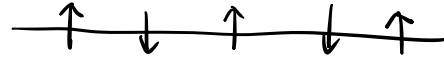
8.1. Haldane chain (1983)

Consider 1D antiferromagnetic spin-S chain:

$$\hat{H} = \sum_i \vec{S}_i \cdot \vec{S}_{i+1}$$



classical configuration:



$$\vec{S}_i \approx (-1)^i S \hat{n}_i + \vec{l}_i$$

$\uparrow$ low E mode $\uparrow$ high E mode

$\int D\vec{l}_i \rightarrow$ involves $\hat{n}_i$

$$Z = \int \left(\prod_i D\vec{n}_i \right) e^{-S[\vec{n}_i]}$$

$$S = \int dx \underbrace{\frac{1}{2g^2} (\partial_\mu \hat{n})^2}_{\text{NLSM}} + i\theta \underbrace{W[\vec{n}]}_{\theta \text{ term}}$$

$$\begin{cases} \theta = 2\pi S \\ W = \int dx \frac{1}{4\pi} \hat{n} \cdot (\partial_t \hat{n} \times \partial_x \hat{n}) \end{cases}$$

Haldane conjecture:

H is $\begin{cases} \text{gapped} \\ \text{gapless} \end{cases}$ if $S \in \begin{cases} \mathbb{Z} \\ \mathbb{Z} + \frac{1}{2} \end{cases}$

If $S \in \mathbb{Z}$, $Z = \int D\hat{n} e^{-S_{\text{NLSM}}} \rightarrow \text{gapped}$

If $S \in \mathbb{Z} + \frac{1}{2}$,

- ① $S = \frac{1}{2}$, Bethe ansatz
- ② $S \in \mathbb{Z} + \frac{1}{2}$, Lieb-Shultz-Mattis thm
- ③ $\int D\hat{n} (-1)^{W[\vec{n}]} e^{-S_{\text{NLSM}}}$

Derivation of θ term.

① Path int for a single spin

$$\vec{S} = \frac{1}{2} \vec{\sigma}$$

$$\text{Consider } \hat{H} = -\vec{n} \cdot \vec{S} = -\frac{1}{2} \vec{n} \cdot \vec{\sigma}$$



$$\vec{n} = (\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta)$$



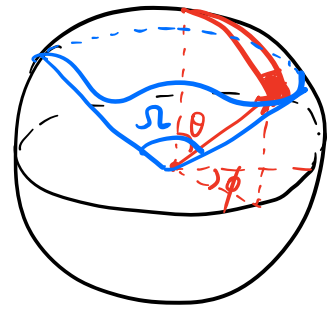
$E = \pm \frac{1}{2}$, ground state is

$$|\vec{n}\rangle = \begin{pmatrix} \cos \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} \end{pmatrix} \Rightarrow \langle \vec{n} | \vec{\sigma}_i | \vec{n} \rangle = n_i$$

For a time evolution $|\vec{n}(t)\rangle$, the Berry phase

$$\begin{aligned} \gamma &= -i \int_0^T dt \langle \vec{n}(t) | \frac{d}{dt} | \vec{n}(t) \rangle \\ &= -i \int_0^T dt \left(\cos \frac{\theta(t)}{2}, e^{-i\phi(t)} \sin \frac{\theta(t)}{2} \right) \begin{pmatrix} \frac{d}{dt} \cos \frac{\theta(t)}{2} \\ \frac{d}{dt} [e^{i\phi(t)} \sin \frac{\theta(t)}{2}] \end{pmatrix} \\ &= -i \int_0^T dt \left[-\frac{\theta'}{2} \sin \frac{\theta}{2} \cos \frac{\theta}{2} + i \phi' \sin^2 \frac{\theta}{2} + \frac{\theta'}{2} \sin \frac{\theta}{2} \cos \frac{\theta}{2} \right] \\ &= \int_0^T dt \frac{1 - \cos \theta(t)}{2} \frac{d\phi(t)}{dt} \\ &= \frac{1}{2} \int d\phi [1 - \cos \theta] \\ &= \frac{1}{2} \int \sin \theta d\theta d\phi \\ &= \frac{1}{2} \Omega[\vec{n}(t)] \end{aligned}$$

↳ solid angle
of trajectory
of $\vec{n}(t)$ on S^2 .



Another SU(2) invariant form of Ω is

$$\Omega = \int_0^1 d\rho \int_0^T dt \vec{n} \cdot (\partial_t \vec{n} \times \partial_\rho \vec{n})$$

$$\text{with } \vec{n}(t, \rho) = \begin{cases} (0, 0, 1), & \text{if } \rho=0 \\ \vec{n}(t), & \text{if } \rho=1 \end{cases}$$

WZ term for one spin

For general spin S :

$$\gamma = S \cdot \Omega[\vec{n}(t)]$$

$$\gamma' = S \cdot (-\Omega') = -S(4\pi - \Omega) \xrightarrow[\substack{\uparrow \\ S \in \frac{1}{2}\mathbb{Z}}]{\text{mod } 2\pi} S\Omega = \gamma$$

single spin :

$$Z = \int D\vec{n}(t) e^{iS \Omega [\vec{n}(t)] + \dots}$$

$\uparrow$
 Wess-Zumino term
 for $(0+1)D$

(2) AF spin chain :



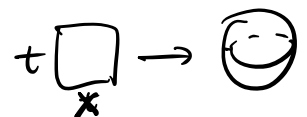
$$Z = \int \left(\prod_i D\vec{n}_i(t) \right) e^{iS \sum_i \Omega [\vec{n}_i(t)] + \dots}$$

$$\vec{n}_i(t) = (-1)^i \vec{m}_i(t) \quad , \quad \text{s.t.} \quad \vec{m}(x,t) := \vec{m}_i(t)$$

is smooth

$$\begin{aligned} S \sum_i \Omega [\vec{n}_i(t)] &= S \sum_i \Omega [(-1)^i m_i(t)] \\ &= S \sum_i (-1)^i \Omega [m_i(t)] \\ &= S \sum_k \left(\Omega [m_{2k}(t)] - \Omega [m_{2k-1}(t)] \right) \\ &= \frac{S}{2} \int_0^L dx \frac{\partial}{\partial x} \Omega [m(x,t)] \\ &= \frac{S}{2} \int_0^L dx \frac{\delta \Omega}{\delta \vec{m}} \cdot \frac{\partial \vec{m}}{\partial x} \\ &= \frac{S}{2} \int dt dx (\vec{m} \times \partial_t \vec{m}) \cdot \partial_x \vec{m} \\ &= 2\pi S \underbrace{\frac{1}{4\pi} \int dt dx \vec{m} \cdot (\partial_t \vec{m} \times \partial_x \vec{m})}_{\substack{= W \\ \text{winding number} \\ (T^2 \rightarrow S^2)}} \end{aligned}$$

$$= 2\pi W$$



$$Z = \int D\vec{n}(x,t) e^{i2\pi S W[\vec{n}(x,t)] + \dots}$$

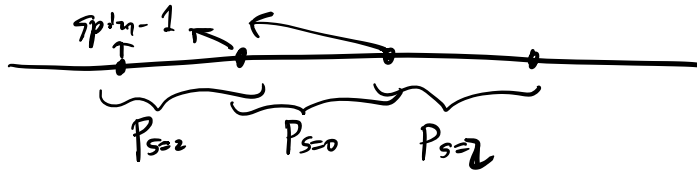
$\downarrow$
 θ term for $(1+1)D$.

8.2. AKLT model

Affleck, Lieb, Kennedy, Tasaki (1987)

- Exactly solvable model for Haldane phase.

$$H_{AKLT} = \sum_i P_{S=2}(\vec{S}_i + \vec{S}_{i+1})$$



spin: $1 \otimes 1 = 0 \oplus 1 \oplus 2$

$$\vec{S}^2 = S(S+1)$$

$$\begin{aligned} (\vec{S}_i + \vec{S}_{i+1})^2 &= \vec{S}_i^2 + \vec{S}_{i+1}^2 + 2 \vec{S}_i \cdot \vec{S}_{i+1} \\ &= 1 \times (1+1) + 1 \times (1+1) + 2 \vec{S}_i \cdot \vec{S}_{i+1} \\ &= 4 + 2 \vec{S}_i \cdot \vec{S}_{i+1} \end{aligned}$$

$$(\vec{S}_i + \vec{S}_{i+1})^2 = S_{tot}(S_{tot}+1) = \begin{cases} 0 & , S_{tot}=0 \\ 2 & , S_{tot}=1 \\ 6 & , S_{tot}=2 \end{cases}$$

$$P_{S=2} = \frac{1}{24} (\vec{S}_i + \vec{S}_{i+1})^2 \cdot [(\vec{S}_i + \vec{S}_{i+1})^2 - 2] = \begin{cases} 0 & , \text{if } S_{tot}=0 \\ 0 & , \text{if } S_{tot}=1 \\ 1 & , \text{if } S_{tot}=2 \end{cases}$$

$$= \frac{1}{24} (4 + 2 \vec{S}_i \cdot \vec{S}_{i+1}) (4 + 2 \vec{S}_i \cdot \vec{S}_{i+1} - 2)$$

$$= \underbrace{\frac{1}{2} \vec{S}_i \cdot \vec{S}_j}_{\text{Heisenberg}} + \underbrace{\frac{1}{6} (\vec{S}_i \cdot \vec{S}_j)^2}_{\text{new terms}} + \frac{1}{3}$$

$$H = \sum_{\langle i,j \rangle} \left[\cos\theta \vec{S}_i \cdot \vec{S}_j + \sin\theta (\vec{S}_i \cdot \vec{S}_j)^2 \right]$$

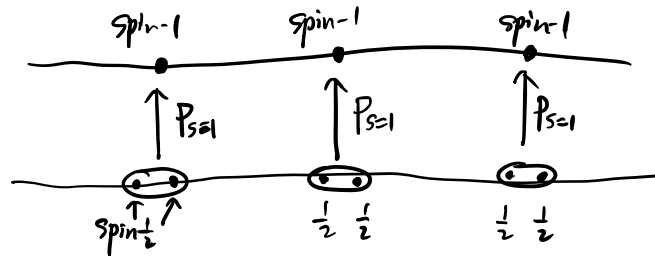
- AKLT state.

$$H = \sum_{\langle i,j \rangle} P_{S=2}(\vec{S}_i + \vec{S}_j) \text{ consists of projectors.}$$

$$\text{But } [P_{S=2}(\vec{S}_{i+1} + \vec{S}_i), P_{S=2}(\vec{S}_i + \vec{S}_{i+1})] \neq 0$$

There is a state $|AKLT\rangle$, s.t. $P_{S=2}(\vec{S}_i + \vec{S}_{i+1}) |AKLT\rangle = 0, \forall i.$

spin- $\frac{1}{2}$ representation:



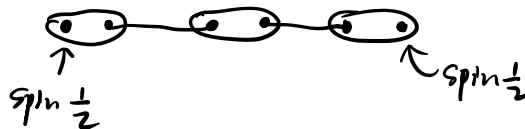
$$|AKLT\rangle = \underbrace{\frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2}}_{\substack{\parallel \\ \frac{1}{2} \otimes \frac{1}{2} \\ \parallel \\ 0 \oplus 1}} \quad \text{where}$$

$$\begin{aligned} \bullet &= \text{spin } \frac{1}{2} \\ \odot &= \left(\frac{1}{2} \otimes \frac{1}{2} \xrightarrow{P_{S=1}} \text{spin } 1 \right) \\ \bullet \text{---} \bullet &= \left(\frac{1}{2} \otimes \frac{1}{2} \xrightarrow{P_{S=0}} \text{spin } 0 \right) \end{aligned}$$

$$\Rightarrow P_{S=2}(\vec{S}_i + \vec{S}_{i+1}) |AKLT\rangle = 0.$$

$\Rightarrow |AKLT\rangle$ is a ground state of H_{AKLT} .

- Properties:
- (1) H_{AKLT} is gapped with GS $|AKLT\rangle$ (Haldane conjecture)
 - (2) H_{AKLT} and $|AKLT\rangle$ preserve $SO(3)_S$, time reversal, lattice inversion symmetries.
 - (3) $|AKLT\rangle$ has spin- $\frac{1}{2}$ edge states for open boundary condition.



$$GSD = 4$$

$AKLT$ is an 1D SPT protected by $SO(3)_S$, time reversal, or lattice inversion symmetries.

- For general symm group G , we can construct AKLT-like state, s.t. the boundary is nontrivial preserving G .

projective rep:

$$U(g) U(h) = \omega_2(g, h) U(gh), \quad \omega_2(g, h) \in U(1)$$

$$(g \cdot h) \cdot k = g \cdot (hk)$$

$$\Rightarrow [U(g) \cdot U(h)] \cdot U(k) = U(g) \cdot [U(h) \cdot U(k)]$$

$$\Rightarrow \omega_2(g, h) U(gh) U(k) = U(g) \cdot \omega_2(h, k) U(hk)$$

$$\Rightarrow \omega_2(g, h) \omega_2(gh, k) \underline{U(ghk)} = \omega_2(h, k) \omega_2(g, hk) \underline{U(ghk)}$$

$$\Rightarrow (d_2 \omega_2)(g, h, k) = \frac{\omega_2(h, k) \omega_2(g, hk)}{\omega_2(gh, k) \omega_2(g, h)} = 1$$

ω_2 is called 2-cocycle of G .

If we do basis transf. $U'(g) := \mu_1(g) U(g)$, for $\mu_1: G \rightarrow U(1)$.

$$\left. \begin{aligned} U'(g) U'(h) &= \omega'_2(g, h) U'(gh) \\ U(g) U(h) &= \omega_2(g, h) U(gh) \end{aligned} \right\}$$

$$\Rightarrow \omega'_2(g, h) = \omega_2(g, h) \cdot \underbrace{\frac{\mu_1(h) \mu_1(g)}{\mu_1(gh)}}_{\equiv: (d_1 \mu_1)(g, h)}$$

$$\Rightarrow \omega'_2 = \omega_2 \cdot (d_1 \mu_1)$$

$\Rightarrow \omega_2$ and ω'_2 give equivalent proj. rep.

Def. $H^2(G, U(1)) = \frac{Z^2(G, U(1))}{B^2(G, U(1))}$ 2nd group cohomology of G , where

$$Z^2(G, U(1)) := \{ \omega_2: G^2 \rightarrow U(1) \mid d_2 \omega_2 = 1 \} \rightarrow \text{2-cocycle}$$

$$B^2(G, U(1)) := \{ d_1 \mu_1 \mid \mu_1: G \rightarrow U(1) \} \rightarrow \text{2-coboundary}$$

$H^2(G, \mathbb{U}(1))$ is an Abelian group :

$$\left. \begin{aligned} U(g) U(h) &= \omega_2(g, h) U(gh) \\ U'(g) &\dots \end{aligned} \right\}$$

$$\Rightarrow [U(g) \otimes U'(g)] [U(h) \otimes U'(h)] = \underbrace{\omega_2(g, h) \omega'_2(g, h)} [U(gh) \otimes U'(gh)]$$

Nontrivial proj. rep must be higher dim.

$$\left. \begin{aligned} U(g) U(h) &= \omega_2(g, h) U(gh) \\ U(g) &\in U(1) \end{aligned} \right\} \Rightarrow \omega_2(g, h) = \frac{U(g) U(h)}{U(gh)} \in B^2(G, \mathbb{U}(1))$$

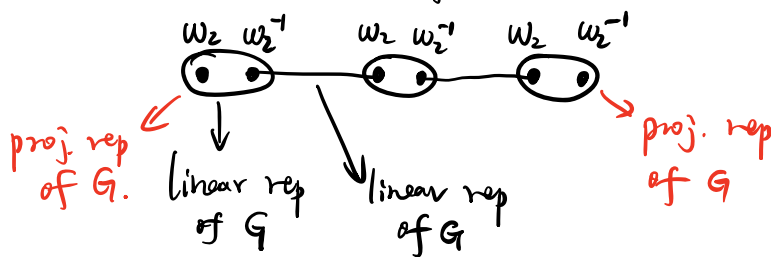
$$\Rightarrow [\omega_2] = [0].$$

Spin $\frac{1}{2}$: $SO(3)$: $\omega_2(\pi, \pi) = -1$

$\mathbb{Z}_2^T$: $\omega_2(T, T) = -1$

projective reps.

AKLT-like state for G :



1D G -SPT is classified (at least) by $H^2(G, \mathbb{U}(1))$.
 ... (at most) ...

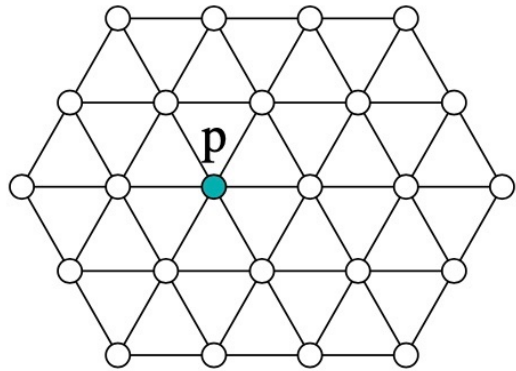
$\Rightarrow$ 1D G -SPT is classified by $H^2(G, \mathbb{U}(1))$.

8.3. Levin-Gu model (2012)

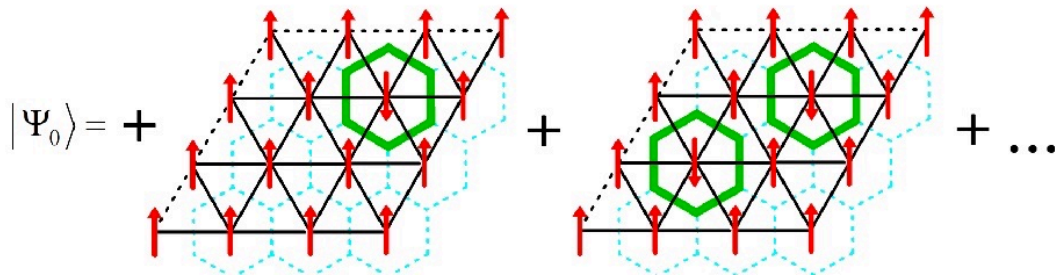
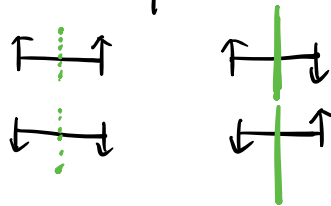
2D SPT protected by $G = \mathbb{Z}_2$.

- Ising paramagnet.

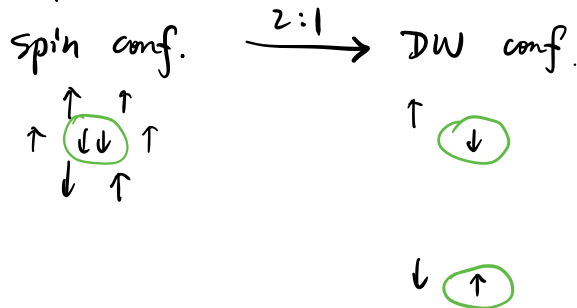
$$\begin{aligned}
 H_0 &= - \sum_p \sigma_p^x \\
 |\psi_0\rangle &= \bigotimes_p |\sigma_p^x = 1\rangle \\
 &= \bigotimes_p \frac{1}{\sqrt{2}} (|\uparrow\rangle_p + |\downarrow\rangle_p) \\
 &\propto \sum_{\{\sigma_p^z = \pm 1\}} |\{\sigma_p^z\}\rangle \\
 &\propto \sum_{\substack{\text{DW conf.} \\ (\text{even, even})}} |\text{DW conf.}\rangle
 \end{aligned}$$



Domain wall picture:



On the plane:

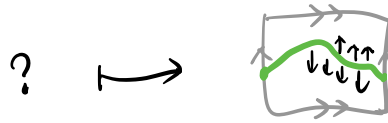


On the torus:

spin conf. $\xrightarrow{2:1}$ DW conf. (even, even)

(odd, even)

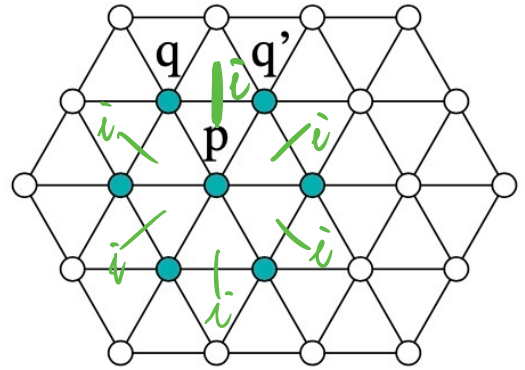
Spin conf. $\xrightarrow{0.1}$ DW conf. (even, odd)
(odd, odd)



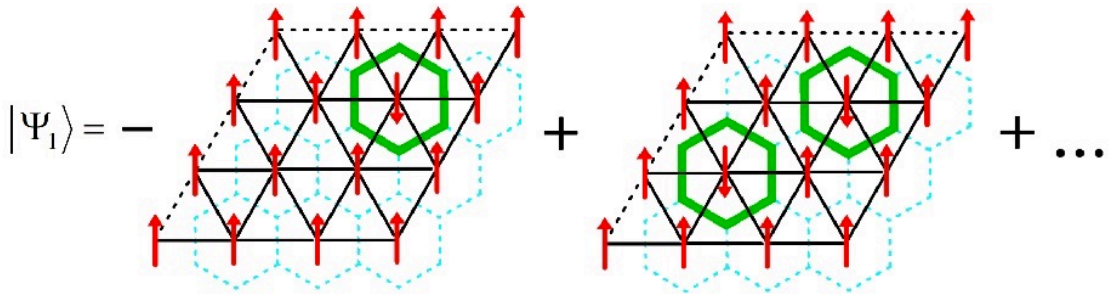
• Levin-Gu state.

$$H_i = - \sum_p B_p$$

$$B_p = - \sigma_p^x \prod_{\langle pq' \rangle} i \frac{1 - \sigma_q^z \sigma_{q'}^z}{2}$$



$$|\Psi_1\rangle = \sum_{\substack{\{\text{DW conf}\} \\ (\text{even, even})}} (-1)^{\#(\text{DW})} |\text{DW conf}\rangle$$



$$- \prod_{\langle pq' \rangle} i \frac{1 - \sigma_q^z \sigma_{q'}^z}{2} = - i^{\#(\text{DW}) \text{ crossing boundary of } \text{hexagon}}$$

$\begin{cases} 1, & \text{if } \sigma_q^z \sigma_{q'}^z = +1 \Leftrightarrow \text{No DW for } \langle qq' \rangle \\ i, & \text{if } \sigma_q^z \sigma_{q'}^z = -1 \Leftrightarrow \text{DW for } \langle qq' \rangle \end{cases}$